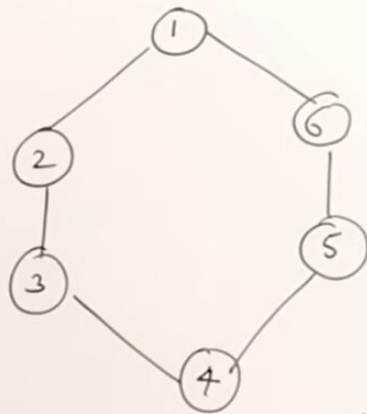


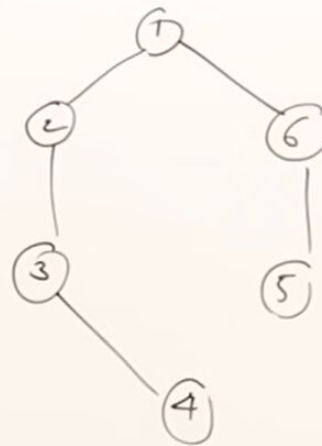
Minimum Cost Spanning Tree



$$G = (V, E)$$

$$V = \{1, 2, 3, 4, 5, 6\}$$

$$E = \{(1,2), (2,3), (3,4), \dots\}$$



$$|V| = n = 6$$

$$|V| - 1 = 5$$

$$S \subseteq G$$

$$S = (V', E')$$

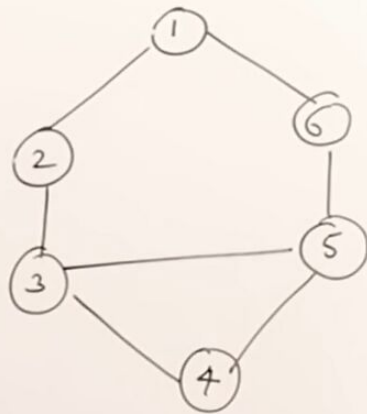
$$V' = V$$

$$|E'| = |V| - 1$$

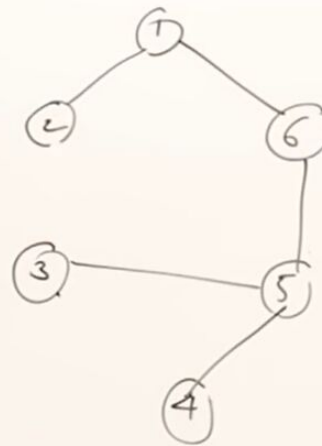
minus one so the number of edges are
number of vertices minus one



Minimum Cost Spanning Tree



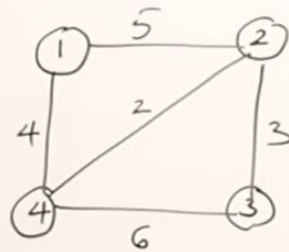
$$|E| = 6$$



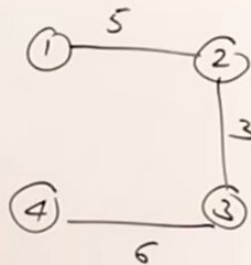
$$7C_5 - 2$$

$$|E| - \frac{|V|-1}{1} - \text{no. of cycles}$$

Minimum Cost Spanning Tree



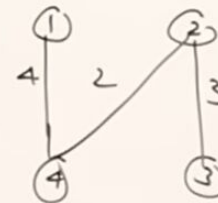
- 1- Prim's
- 2- Kruskal's



Cost = 14



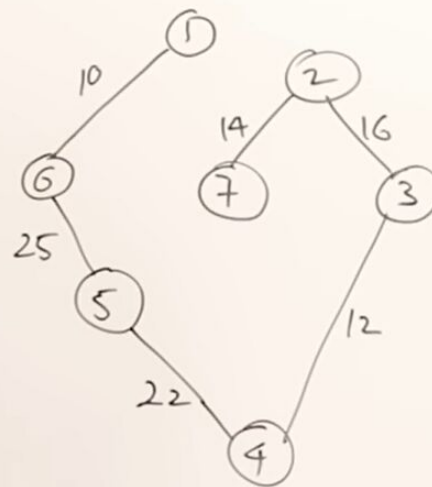
Cost = 13



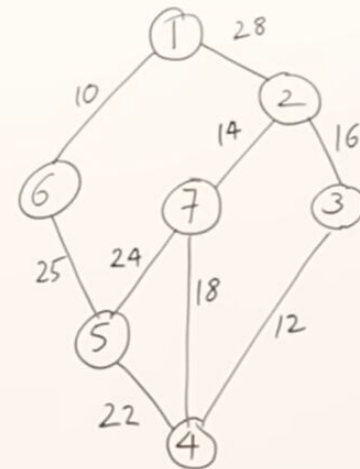
Cost = 9

Minimum Cost Spanning Tree

Prim's



$$\text{Cost} = 99$$

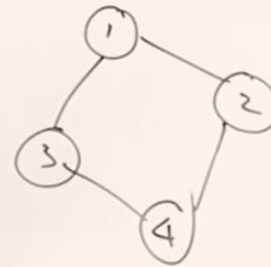
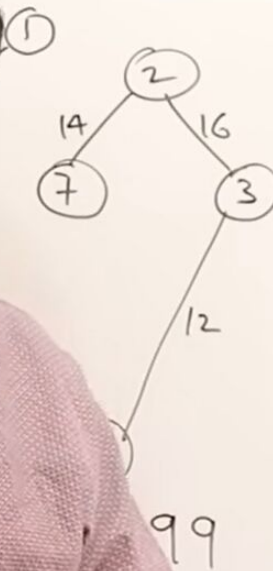


select the smallest one then always
select the connected smallest



Minimum Cost Spanning Tree

Prim's

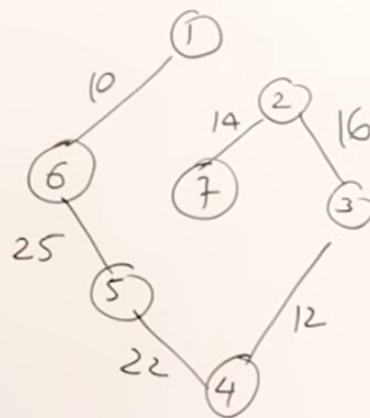


graph can Prim find out the minimum cost spanning tree for this one no



Minimum Cost Spanning Tree

Kruskal's



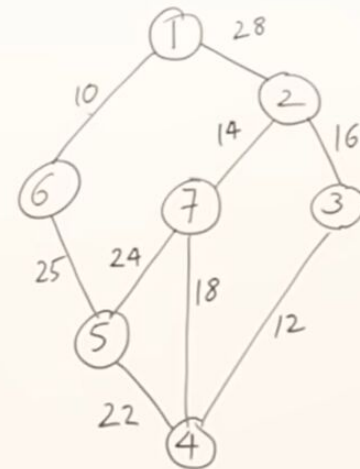
Cost = 99

$|E|$

$|V| - 1$

$O(|V||E|)$

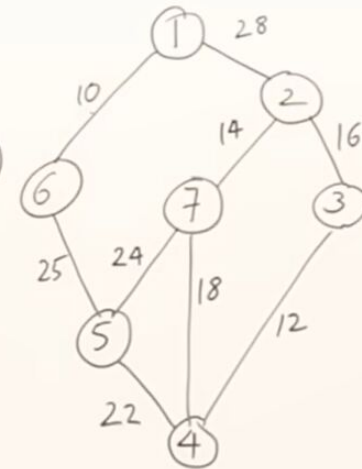
$O(n \cdot e) = O(n^2)$



Minimum Cost Spanning Tree

Kruskal's

Min Heap
 $\Theta(n \log n)$



$|V| - 1$

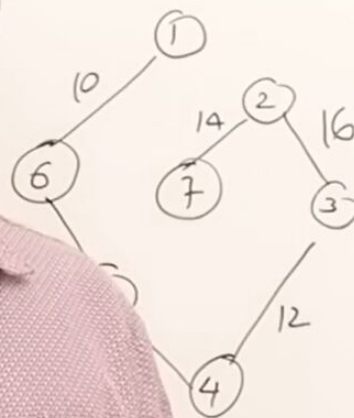
$$\Theta(|V| |E|)$$
$$\Theta(n \cdot e) = \Theta(n^2)$$

minus one time so n so by using Min he
the time time complexity



Minimum Cost Spanning Tree

Kruskal's



Cost = 99

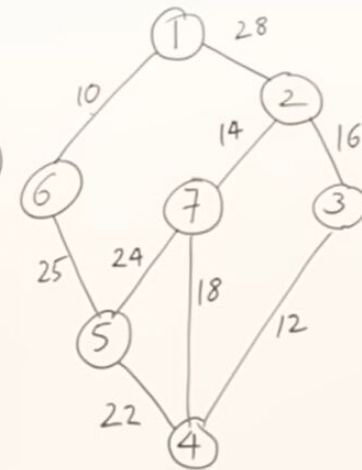
Min Heap
 $O(n \log n)$

$|E|$

$|V| - 1$

$O(|V| |E|)$

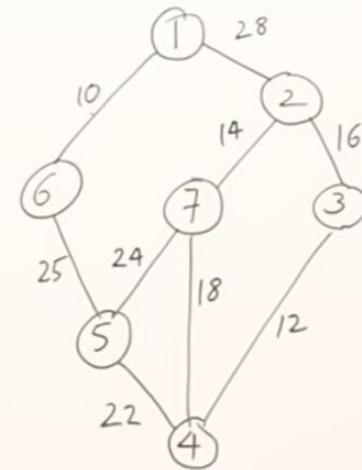
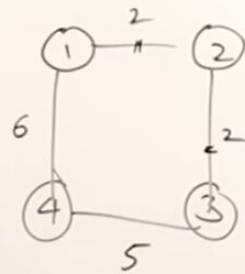
$O(n^2) = O(n^2)$



the time time complexity of crysal
algorithm will be $n \log n$



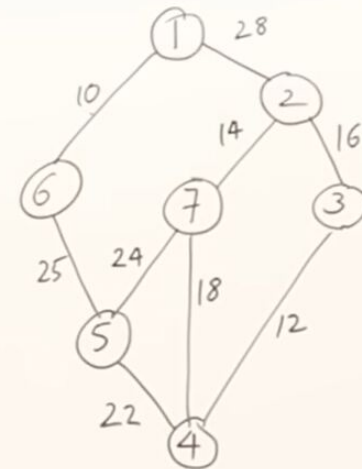
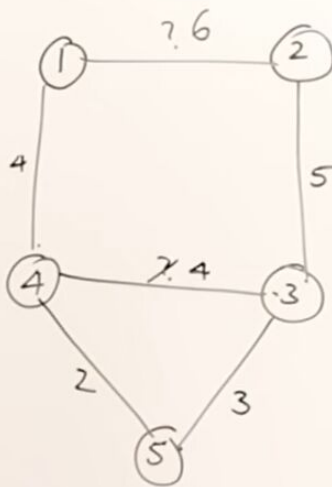
Minimum Cost Spanning Tree



the entire graph so crissal algo then may work for non-connected graphs



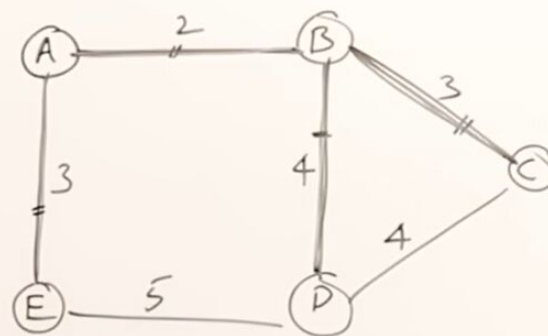
Minimum Cost Spanning Tree



six so if any missing edges are there
you can find out these



Minimum Cost Spanning Tree

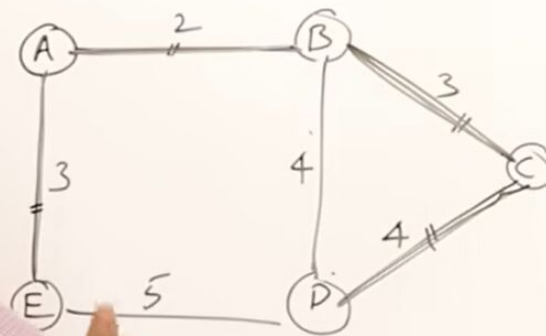


$$2 + 3 + 3 + 4 = 12$$

have exactly one spanning tree only there



Minimum Cost Spanning Tree



$$+ 3 + 3 + 4 = 12$$

what is the cost cost is 12 so cost is only one the

